1 Design Space

1.1 Conceptual Design Space

- Refers to the full range of possibilities for addressing identified problems
- Infinite in scope
- Examples: Digital vs. physical, manual vs. automatic, input/output modalities

1.2 Realistic Design Space

- Mapping of one or two dimensions of the design
- Organizes and suggests possibilities
- Tangible representations of design ideas

2 Making Design Concrete

- Design process involves lots of cognitive resources, information, and unknowns
- Use tangible representations: sketches, prototypes, written notes
- Facilitates managing complexity

3 What is a Prototype?

- Limited representation of an envisioned design
- Allows interaction and exploration of suitability
- Examples: Paper prototyping, screen sketches, storyboards, cardboard mock-ups, Powerpoint slides, videos, software prototypes

4 Why Prototype?

- Evaluation and feedback are central
- Stakeholders can interact with prototypes
- Enhances team communication
- Encourages reflection and design choices
- Answers questions and supports decision-making

5 Low-fidelity Prototyping

- Uses media unlike the final form (e.g., paper, cardboard)
- Quick, cheap, and easily changeable
- Example: Card-based prototypes with index cards

6 High-fidelity Prototyping

- Prototype resembles the final system
- User-driven and interactive
- No back-end functionality
- Common environments: HTML, Flash, Powerpoint
- Includes mockups resembling the final interface

7 Advantages and Disadvantages

- **Low-fidelity**: Cheaper, evaluates concepts, communicates, addresses screen layouts, proof-of-concept
- **High-fidelity**: Functional, defines navigation, user-driven, living specification

8 Disadvantages

- **Low-fidelity**: Limited error checking, poor coding specs, facilitator-driven, limited utility after requirements, navigation issues
- **High-fidelity**: Expensive, time-consuming, inefficient for PoCs, not effective for requirements gathering